

PY1010 — Physics for Biological and Chemical Sciences

Summer 2025 Examination · Paper 1 · Full Marking Scheme

Module	PY1010 — Physics for Biological and Chemical Sciences
Internal examiners	Dr M. Kennedy, Prof. A. Ruth
Institution	University College Cork
Session	Summer 2025, Paper 1
Duration	3 hours (+ 15 minutes reading time)
Structure	Six questions, each given as an either/or choice (A or B). Answer one option of each of the six questions and all parts of that option. Marks for each part are shown in square brackets.
Materials	Non-programmable calculator permitted; constants on the paper's “useful information” page

How to use this scheme. Every answer is worked out in small numbered steps, with a “Rules you need” box listing the formulas used. Nothing is skipped, so you can see where every number comes from: every unit conversion, every substitution, every check. Read the answer first, then follow the steps.

Coverage note. This scheme works through **both options (A and B) of all six questions** in full. On the day you only choose one option per question, but working both lets you pick the one you're strongest on.

Source basis. Every answer is built **only** from the PY1010 lecture material given to students: the lecture slides (mechanics, fluids, thermodynamics) and the handwritten lecture notes (gas laws, electricity, magnetism, waves and optics). No textbook or outside material is used. Where the paper asks for something the lecture notes don't state explicitly, a highlighted note says so at that part (Q4(B)(a) and Q6(A)(d)).

Own-figure rule. Where a part carries forward a number from an earlier part, an answer that uses the correct method on an earlier wrong value still earns the method marks for the later part.

Numbers. $g = 9.81 \text{ m s}^{-2}$ and the other constants come from the paper's useful-information page. Intermediate values are kept unrounded; final answers are given to 2–3 significant figures. Answers differing only through rounding (e.g. using $g = 9.8$) get full marks.

Question 1 · Option A

Speed vs velocity, area, density and mass flow of blood — 10 marks

Given: Blood: average speed 1.4 km/hr, density 1060 kg/m^3 ; aorta radius 1.2 cm; 45% of blood is red blood cells; mass of one red blood cell $27 \times 10^{-15} \text{ kg}$.

1(A)(a)

[1 mark]

Express the average speed of blood in m/s.

ANSWER

$$v = 0.39 \text{ m/s (0.3889 m/s)}$$

RULES YOU NEED

- $1 \text{ km} = 1000 \text{ m}$
- $1 \text{ hr} = 60 \text{ min} \times 60 \text{ s} = 3600 \text{ s}$
- To convert a unit, multiply by a fraction equal to 1 (e.g. $1000 \text{ m} / 1 \text{ km}$) so the old unit cancels.

WORKING — STEP BY STEP

1 Write down what you are given

The speed is 1.4 km/hr. We want it in m/s, so both the kilometres and the hours have to change.

2 Change kilometres to metres

There are 1000 m in 1 km, so multiply by 1000:

$$1.4 \text{ km} = 1.4 \times 1000 \text{ m} = 1400 \text{ m}$$

3 Change hours to seconds

One hour is 60 minutes, and each minute is 60 seconds:

$$1 \text{ hr} = 60 \times 60 \text{ s} = 3600 \text{ s}$$

4 Divide the distance by the time

So 1.4 km/hr means 1400 m travelled in 3600 s:

$$v = 1400 \text{ m} \div 3600 \text{ s} = 0.3889 \text{ m/s}$$

5 Round the final answer

Round to 2 significant figures for the answer line, but keep 0.3889 in your calculator for later parts:

$$v \approx 0.39 \text{ m/s}$$

EXAM TECHNIQUE

Quick shortcut: to go from km/hr to m/s, divide by 3.6 (because $1000 \div 3600 = 1/3.6$). Check: $1.4 \div 3.6 = 0.389 \checkmark$.

Marks**Awarded for****1**Correct conversion to 0.39 m/s **with unit**.

1(A)(b)**[2 marks]**

What is the difference between speed and velocity?

ANSWER

Speed is a scalar (distance/time); velocity is a vector (displacement/time) with magnitude and direction.

RULES YOU NEED

- Average speed = total distance ÷ total time
- Average velocity = displacement ÷ time, $v = \Delta x / \Delta t$
- Displacement and velocity are **vectors** (magnitude + direction).

WORKING — STEP BY STEP**1 What is speed?**

Speed = total distance travelled ÷ total time taken.

It is just a size (a number with a unit, e.g. 3 m/s). It is a **scalar**: it says nothing about which way you are going.

2 What is velocity?

Velocity = displacement ÷ time, $v = \Delta x / \Delta t$.

Displacement is the change in position (final position – start position), which has a direction. So velocity has a **size and a direction**. It is a **vector**.

3 See the difference with an example

A car drives one full lap of a 5 km track in 5 minutes and stops where it started.

Its average speed is 5 km in 5 min, which is not zero.

Its displacement is zero (it ended where it started), so its average velocity is **zero**.

4 Write the answer in one line

Speed tells you how fast; velocity tells you how fast **and which way**.

Marks	Awarded for
1	Speed defined as distance/time with no direction (scalar).
1	Velocity defined as displacement/time with magnitude and direction (vector).

1(A)(c)**[2 marks]**

Calculate the cross sectional area of a typical aorta, giving your answer in m^2 .

ANSWER

$$A = 4.5 \times 10^{-4} \text{ m}^2 \text{ (} 4.52 \times 10^{-4} \text{ m}^2 \text{)}$$

RULES YOU NEED

- Area of a circle: $A = \pi r^2$
- $1 \text{ cm} = 10^{-2} \text{ m}$ (centi = 10^{-2})

WORKING — STEP BY STEP**1 The cross-section is a circle**

If you slice across the aorta, the cut face is a circle of radius $r = 1.2 \text{ cm}$. Its area is $A = \pi r^2$.

2 Change the radius to metres first

The answer must be in m^2 , so put r in metres before squaring:

$$r = 1.2 \text{ cm} = 1.2 \times 10^{-2} \text{ m} = 0.012 \text{ m}$$

3 Square the radius

$$r^2 = (0.012 \text{ m})^2 = 0.000144 \text{ m}^2 = 1.44 \times 10^{-4} \text{ m}^2$$

4 Multiply by π

$$A = \pi \times 1.44 \times 10^{-4} \text{ m}^2 = 4.52 \times 10^{-4} \text{ m}^2$$

EXAM TECHNIQUE

Convert to metres *before* you square. If you square 1.2 cm first you get 1.44 cm^2 , and then you must remember that $1 \text{ cm}^2 = 10^{-4} \text{ m}^2$ (not 10^{-2}).

Marks	Awarded for
1	Correct conversion of r to metres and use of $A = \pi r^2$.
1	Correct answer in m^2 .

1(A)(d)**[3 marks]**

Calculate the mass of blood which leaves an aorta in 1 second.

ANSWER **$m \approx 0.19 \text{ kg of blood per second (0.186 kg)}$** **RULES YOU NEED**

- Distance travelled at constant speed: $x = v t$
- Volume of a cylinder: $V = A \times \text{length}$
- Density: $\rho = m / V$, so $m = \rho V$

WORKING — STEP BY STEP**1 Picture what leaves in 1 second**

In 1 s the blood moves forward a distance x . All the blood that leaves fills a cylinder: its end is the aorta's circle (area A) and its length is x .

2 How far does the blood move in 1 s?

Use the speed from part (a):

$$x = v t = 0.3889 \text{ m/s} \times 1 \text{ s} = 0.3889 \text{ m}$$

3 Find the volume of that cylinder

Use the area from part (c):

$$V = A \times x = 4.524 \times 10^{-4} \text{ m}^2 \times 0.3889 \text{ m} = 1.759 \times 10^{-4} \text{ m}^3$$

4 Turn volume into mass using density

Density tells you how many kg are in each m^3 . Rearranging $\rho = m/V$ gives $m = \rho V$:

$$m = 1060 \text{ kg/m}^3 \times 1.759 \times 10^{-4} \text{ m}^3 = 0.1865 \text{ kg}$$

5 State the answer with a unit

$$m \approx 0.19 \text{ kg of blood every second}$$

Sense check: 0.19 kg is about 180 ml of blood per second, a sensible amount for the body's main artery.

Marks	Awarded for
1	Distance travelled in 1 s ($x = vt$) or volume per second ($V = Av$).
1	Correct volume of blood.
1	Correct mass via $m = \rho V$, with unit.

Study-Uni

1(A)(e)**[2 marks]**

Calculate the number of red blood cells which leave an aorta in 1 second.

ANSWER

$N \approx 3.1 \times 10^{12}$ red blood cells per second

RULES YOU NEED

- 45% of something = $0.45 \times \text{it}$
- Number of items = total mass $\div$ mass of one item

WORKING — STEP BY STEP**1 Find the mass that is red blood cells**

45% of the blood is red blood cells. Take 45% of the mass from part (d):

$$m_{\text{RBC}} = 0.45 \times 0.1865 \text{ kg} = 0.08392 \text{ kg}$$

2 Divide by the mass of one cell

Each red blood cell has mass $27 \times 10^{-15} \text{ kg}$. The number of cells is the total mass divided by the mass of one:

$$N = 0.08392 \text{ kg} \div (27 \times 10^{-15} \text{ kg})$$

3 Do the division carefully with powers of ten

$0.08392 \div 27 = 0.003108$, and dividing by 10^{-15} is the same as multiplying by 10^{15} :

$$N = 0.003108 \times 10^{15} = 3.1 \times 10^{12}$$

EXAM TECHNIQUE

The question doesn't say whether the 45% is by mass or by volume. Taking it as 45% of the mass from (d) is the natural reading. If you take it by volume, say you assume the red cells have the same density as whole blood; the answer is the same.

Marks	Awarded for
1	45% applied to the mass from (d).
1	Correct division, giving $\approx 3.1 \times 10^{12}$ (own-figure from (d) accepted).

Question 1(A) Total: 10 Marks

Question 1 · Option B

Acceleration and force, glacier kinematics, friction — 10 marks

Given: Jakobshavn Glacier velocity (2013): 17 km/year.

1(B)(a)

[2 marks]

Define and give the SI units for (i) acceleration, and (ii) force.

ANSWER

$$a = \Delta v / \Delta t \text{ [m s}^{-2}\text{]}; F = ma \text{ [N = kg m s}^{-2}\text{]}$$

RULES YOU NEED

- $a = \Delta v / \Delta t$ (change in velocity ÷ time taken)
- Newton's second law: $F = dp/dt = ma$
- $1 \text{ N} = 1 \text{ kg m s}^{-2}$

WORKING — STEP BY STEP

1 Acceleration

Acceleration is how quickly velocity changes: the change in velocity divided by the time the change takes.

$$a = \Delta v / \Delta t = (v_2 - v_1) / (t_2 - t_1)$$

Units: $(\text{m/s}) \div \text{s} = \text{m s}^{-2}$. Acceleration is a vector.

2 Force

Newton's second law: the net force equals the rate of change of momentum. For a fixed mass this is mass × acceleration:

$$F = dp/dt = m a$$

Units: $\text{kg} \times \text{m s}^{-2} = \text{kg m s}^{-2}$, which is called the **newton (N)**. One newton gives a 1 kg mass an acceleration of 1 m s^{-2} .

Marks	Awarded for
1	Acceleration defined as $\Delta v / \Delta t$ with unit m s^{-2} .
1	Force defined ($F = ma$ / rate of change of momentum) with unit N or kg m s^{-2} .

1(B)(b)**[2 marks]**

Do force and acceleration always point in the same direction? Explain.

ANSWER

Yes for the net force: a is always along F_{net} ($F = ma$, $m > 0$). No for individual forces, e.g. friction opposes the acceleration of a pulled box.

RULES YOU NEED

- Newton's second law: the rate of change of momentum is proportional to the **net** force and in the **same direction**.
- $F_{\text{net}} = m a$, and mass m is always positive.

WORKING — STEP BY STEP**1 Look at $F = ma$**

Mass is a positive number. Multiplying a vector by a positive number never changes its direction. So the acceleration a always points the same way as the **net** force F_{net} .

2 But not every single force

An object can have several forces on it. Only their total (the net force) sets the acceleration.

Example from the lectures: a box pulled across a table with 101 N while friction pulls back with 78.4 N. Friction points backwards, but the box accelerates forwards, because the net force ($101 - 78.4 = 22.6$ N) is forwards.

3 And not always the velocity either

A car braking moves forwards but accelerates backwards. Its acceleration points along the net force (the braking force), opposite to its motion.

4 Conclusion

Yes for the net force; no for each individual force.

Marks	Awarded for
1	States acceleration is always in the direction of the net force, justified with $F = ma$.
1	Distinguishes individual forces (or velocity) from the net force, with an example.

1(B)(c)**[1 mark]**

Assuming the glacier's speed was constant for the 90 days of summer, calculate the total distance travelled, in metres.

ANSWER

$$x \approx 4.2 \times 10^3 \text{ m (4192 m)}$$

RULES YOU NEED

- Constant velocity: distance = velocity $\times$ time
- 1 year = 365 days, 1 km = 1000 m

WORKING — STEP BY STEP**1 Put the speed into metres per day**

17 km/year = 17 000 m per 365 days:

$$v = 17\,000 \text{ m} \div 365 \text{ days} = 46.58 \text{ m/day}$$

2 Multiply by the time

$$x = v t = 46.58 \text{ m/day} \times 90 \text{ days} = 4192 \text{ m}$$

3 Check with a fraction of a year

90 days is 90/365 of a year, so the glacier moves 90/365 of 17 km:

$$17\,000 \text{ m} \times 90/365 = 4192 \text{ m} \checkmark$$

Marks**Awarded for****1**

Correct distance in metres.

1(B)(d)**[2 marks]**

The glacier decelerates at a constant rate and stops after 30 days. Calculate its deceleration in SI units.

ANSWER

$$a = -2.1 \times 10^{-10} \text{ m s}^{-2} \text{ (deceleration of } 2.1 \times 10^{-10} \text{ m s}^{-2}\text{)}$$

RULES YOU NEED

- $v = v_0 + at$, so $a = (v - v_0)/t$
- SI units: metres and seconds
- 1 day = $24 \times 3600 \text{ s} = 86\,400 \text{ s}$; 1 year = $365 \times 86\,400 \text{ s}$

WORKING — STEP BY STEP**1 Change the starting velocity to m/s**

One year in seconds: $365 \times 24 \times 3600 = 31\,536\,000 \text{ s}$.

$$v_0 = 17\,000 \text{ m} \div 31\,536\,000 \text{ s} = 5.39 \times 10^{-4} \text{ m/s}$$

2 Change the time to seconds

$$t = 30 \text{ days} \times 86\,400 \text{ s/day} = 2.592 \times 10^6 \text{ s}$$

3 Use $v = v_0 + at$

The glacier ends at rest, so $v = 0$. Rearrange for a :

$$a = (v - v_0) / t = (0 - 5.39 \times 10^{-4}) \div 2.592 \times 10^6$$

4 Calculate and interpret the sign

$$a = -2.08 \times 10^{-10} \text{ m s}^{-2}$$

The minus sign means it is slowing down: a deceleration of $2.08 \times 10^{-10} \text{ m s}^{-2}$.

Marks	Awarded for
1	Velocity and time both converted to SI (m s^{-1} , s).
1	Correct deceleration with unit m s^{-2} .

1(B)(e)**[3 marks]**

The deceleration was caused by an increase in the coefficient of friction between glacier and ground. Calculate the coefficient of friction required to provide this deceleration.

ANSWER

$$\mu_k \approx 2.1 \times 10^{-11} \text{ (unitless)}$$

RULES YOU NEED

- Kinetic friction: $F_{\text{friction}} = \mu_k N$
- On flat ground: $N = mg$
- Newton's second law: $F = ma$
- μ has no unit

WORKING — STEP BY STEP**1 Which force stops the glacier?**

The glacier is moving, so kinetic friction acts on it, backwards. Friction is the only horizontal force, so it is the net force.

2 Write the friction force

On flat ground the normal force equals the weight, $N = mg$:

$$F_{\text{friction}} = \mu_k N = \mu_k m g$$

3 Set friction equal to ma

$$\mu_k m g = m |a|$$

The mass m is on both sides, so it cancels. We don't need the glacier's mass:

$$\mu_k = |a| / g$$

4 Put the numbers in

$$\mu_k = 2.08 \times 10^{-10} \div 9.81 = 2.1 \times 10^{-11}$$

μ is force $\div$ force, so it has **no unit**.

Marks	Awarded for
1	Friction written as μN with $N = mg$.
1	Friction set equal to ma (Newton's 2nd law) so that m cancels.
1	Correct $\mu \approx 2.1 \times 10^{-11}$, no unit (own-figure from (d) accepted).

Question 1(B) Total: 10 Marks

Question 2 · Option A

Uniform circular motion, the centrifuge, kinetic energy — 10 marks

Given: Centrifuge spins at 55 rotations per second; $F_c = m\omega^2 r$ is given.

2(A)(a)

[2 marks]

An object in uniform circular motion has a constantly changing velocity. By referencing Newton's second law, explain why it must be experiencing a force.

ANSWER

Changing direction $\Rightarrow$ changing velocity $\Rightarrow$ acceleration $\Rightarrow$ by $F = ma$ there must be a net (centripetal) force.

RULES YOU NEED

- Velocity is a vector: changing its direction changes it.
- $a = \Delta v / \Delta t$
- Newton's second law: $F_{\text{net}} = ma$
- Newton's first law: no net force $\rightarrow$ straight line at constant speed

WORKING — STEP BY STEP

1 Why is the velocity changing?

In uniform circular motion the speed stays the same, but the **direction** keeps turning as the object goes round. Velocity includes direction, so the velocity is changing all the time.

2 A changing velocity means acceleration

Acceleration is the rate of change of velocity, $a = \Delta v / \Delta t$. The velocity changes, so $a \neq 0$: the object is always accelerating (towards the centre).

3 Apply Newton's second law

$F_{\text{net}} = ma$. If $a \neq 0$, then $F_{\text{net}} \neq 0$. So there must be a net force, pointing towards the centre: the **centripetal force**.

4 Check with Newton's first law

With no force, the object would carry on in a straight line. Something must keep pulling it round, which confirms a force is needed.

Marks

Awarded for

1 Velocity changes because its direction changes, so the object accelerates.

1 Links acceleration to a net force through $F = ma$.

2(A)(b)**[1 mark]**

How long does it take the centrifuge to complete 1 full rotation? Give your answer in seconds.

ANSWER

$T = 0.018 \text{ s (18.2 ms)}$

RULES YOU NEED

- Frequency f = rotations per second (Hz)
- Period T = time for one rotation = $1/f$

WORKING — STEP BY STEP**1 Identify the frequency**

55 rotations per second is the frequency: $f = 55 \text{ s}^{-1}$ (55 Hz).

2 The period is $1 \div \text{frequency}$

If 55 rotations take 1 second, one rotation takes $1/55$ of a second:

$$T = 1 / f = 1 / 55 = 0.0182 \text{ s}$$

Marks**Awarded for****1** $T = 1/f = 0.018 \text{ s.}$

2(A)(c)**[2 marks]**

What is the acceleration felt by a sample 8 cm from the axis of rotation?

ANSWER

$a \approx 9.6 \times 10^3 \text{ m s}^{-2}$ ($\approx 970 \text{ g}$), directed towards the centre

RULES YOU NEED

- Angular velocity $\omega = 2\pi/T = 2\pi f$ (rad/s)
- $F_c = m\omega^2 r$ (given) and $F = ma$, so $a = \omega^2 r$

WORKING — STEP BY STEP**1 Get ω in radians per second**

One rotation is 2π radians, and there are 55 rotations each second:

$$\omega = 2\pi f = 2\pi \times 55 = 345.6 \text{ rad/s}$$

2 Find a formula for the acceleration

The question gives $F_c = m\omega^2 r$. Newton's second law says $a = F/m$, so divide by m :

$$a = F_c / m = \omega^2 r$$

3 Put r in metres

$$r = 8 \text{ cm} = 0.08 \text{ m}$$

4 Calculate

$$a = (345.6)^2 \times 0.08 = 119\,400 \times 0.08 = 9.55 \times 10^3 \text{ m s}^{-2}$$

5 Compare with g to get a feel for it

$$9554 \div 9.81 \approx 970$$

So the sample feels about 970 times the pull of gravity, directed towards the axis.

EXAM TECHNIQUE

Don't use $\omega = 55$. That is the frequency in rotations per second. The formula needs ω in rad/s, so multiply by 2π first.

Marks	Awarded for
1	Correct $\omega = 2\pi f$ (in rad s^{-1}) and $a = \omega^2 r$.
1	Correct value with unit.

Study-Uni

2(A)(d)**[3 marks]**

If the tube broke instantly, how far would the sample fly in 1 second? Ignore gravity.

ANSWER

The sample flies ≈ 28 m (27.6 m) in a straight line along the tangent

RULES YOU NEED

- Speed around a circle: $v = \omega r$
- Newton's first law: no force $\rightarrow$ straight line at constant velocity
- Distance = $v t$

WORKING — STEP BY STEP**1 Find the sample's speed just before the break**

It is moving around a circle of radius 0.08 m at $\omega = 345.6$ rad/s:

$$v = \omega r = 345.6 \times 0.08 = 27.65 \text{ m/s}$$

2 What happens when the tube breaks?

The centripetal force disappears, and gravity is ignored, so there is **no net force**. By Newton's first law the sample carries on at constant velocity in a **straight line**.

3 Which direction?

Along the tangent to the circle at the point where it was released, i.e. in the direction it was moving at that instant (not outwards from the centre).

4 Distance in 1 second

$$x = v t = 27.65 \text{ m/s} \times 1 \text{ s} = 27.6 \text{ m}$$

Marks	Awarded for
1	Tangential speed $v = \omega r$.
1	No net force, so straight-line motion at constant speed (Newton's 1st law).
1	Correct distance ≈ 27.6 m (own-figure from (c) accepted).

2(A)(e)**[2 marks]**

If the sample has a mass of 5 g, calculate its kinetic energy in Joules.

ANSWER

$$E_{\text{kin}} \approx 1.9 \text{ J}$$

RULES YOU NEED

- $E_{\text{kin}} = \frac{1}{2} m v^2$
- $1 \text{ g} = 10^{-3} \text{ kg}$
- $1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2}$

WORKING — STEP BY STEP**1 Convert the mass to kg**

$$m = 5 \text{ g} = 5 \times 10^{-3} \text{ kg} = 0.005 \text{ kg}$$

2 Square the speed from (d)

$$v^2 = (27.65)^2 = 764.4 \text{ m}^2 \text{ s}^{-2}$$

3 Apply the kinetic energy formula

$$E_{\text{kin}} = \frac{1}{2} \times 0.005 \times 764.4 = 1.91 \text{ J}$$

Marks	Awarded for
1	Mass in kg and $E_{\text{kin}} = \frac{1}{2}mv^2$ with the speed from (d).
1	Correct value in J.

Question 2(A) Total: 10 Marks

Question 2 · Option B

Conservation of energy, work done by air resistance — 10 marks

Given: A pine cone of mass 0.14 kg falls 16 m and lands at 13 m/s.

2(B)(a)

[1 mark]

State the law of conservation of energy and the precondition for its applicability.

ANSWER

$E_{\text{pot}} + E_{\text{kin}} = \text{constant}$, provided only conservative forces act

RULES YOU NEED

- $E_{\text{total}} = E_{\text{pot}} + E_{\text{kin}}$
- Conservative forces (gravity, springs): work depends only on start and end points
- Dissipative forces (friction): $E_{\text{initial}} - E_{\text{final}} = W_{\text{friction}}$

WORKING — STEP BY STEP

1 The law

The total mechanical energy, potential + kinetic, stays constant:

$$E_{\text{pot}} + E_{\text{kin}} = \text{constant}$$

2 The precondition

This only holds when **only conservative forces** act (e.g. gravity). If friction or air resistance acts, some energy is lost as heat, and the total drops.

Marks

Awarded for

1

Statement **and** the conservative-forces-only condition.

2(B)(b)**[2 marks]**

Assuming no air friction, what velocity would the pine cone have immediately before touching the ground?

ANSWER

$$v \approx 17.7 \text{ m s}^{-1}$$

RULES YOU NEED

- $E_{\text{pot}} = mgh$
- $E_{\text{kin}} = \frac{1}{2}mv^2$
- No friction: PE lost = KE gained, so $v = \sqrt{2gh}$

WORKING — STEP BY STEP**1 Energy at the top**

The cone starts at rest 16 m up, so all its energy is potential: mgh .

2 Energy at the bottom

At the ground $h = 0$, so all of it has become kinetic energy: $\frac{1}{2}mv^2$.

3 Set them equal

$$m g h = \frac{1}{2} m v^2$$

Cancel m from both sides and rearrange:

$$v = \sqrt{2 g h}$$

4 Put the numbers in

$$v = \sqrt{2 \times 9.81 \times 16} = \sqrt{313.9} = 17.7 \text{ m/s}$$

Marks	Awarded for
1	Energy conservation (or free-fall equation) set up correctly.
1	Correct $v \approx 17.7 \text{ m s}^{-1}$.

2(B)(c)**[1 mark]**

Does air resistance do positive, negative, or no work on the pine cone? Briefly explain.

ANSWER**Negative work: the force opposes the displacement****RULES YOU NEED**

- Work sign: positive if force is along the displacement, negative if opposite, zero if perpendicular.
- Friction always points opposite to the motion.

WORKING — STEP BY STEP**1 Which way do the force and motion point?**

The cone moves **down**. Air resistance pushes **up**, against the motion.

2 Apply the sign rule

Force opposite to displacement → **negative** work.

3 Check against the numbers

The cone lands at 13 m/s, slower than the 17.7 m/s it would have with no air. Energy was removed, which is what negative work means.

Marks**Awarded for****1**

“Negative”, with the reason (force opposite to displacement / energy removed).

2(B)(d)**[3 marks]**

How much work is done on the pine cone by air resistance?

ANSWER

$$W_{\text{air}} \approx -10.1 \text{ J (10.1 J of mechanical energy dissipated)}$$

RULES YOU NEED

- $E_{\text{pot}} = mgh$, $E_{\text{kin}} = \frac{1}{2}mv^2$
- With a dissipative force: $W_{\text{air}} = E_{\text{final}} - E_{\text{initial}}$

WORKING — STEP BY STEP**1 Energy at the top (at rest, $h = 16 \text{ m}$)**

$$E_{\text{initial}} = m g h = 0.14 \times 9.81 \times 16 = 21.97 \text{ J}$$

2 Energy at the bottom ($h = 0$, $v = 13 \text{ m/s}$)

$$E_{\text{final}} = \frac{1}{2} m v^2 = \frac{1}{2} \times 0.14 \times 13^2 = \frac{1}{2} \times 0.14 \times 169 = 11.83 \text{ J}$$

3 The difference is the work done by air resistance

$$W_{\text{air}} = E_{\text{final}} - E_{\text{initial}} = 11.83 - 21.97 = -10.1 \text{ J}$$

The minus sign agrees with part (c): air resistance did negative work and removed about 10 J (turned into heat).

Marks	Awarded for
1	Initial potential energy $mgh = 22.0 \text{ J}$.
1	Final kinetic energy $\frac{1}{2}mv^2 = 11.8 \text{ J}$.
1	Work = difference, -10.1 J , with the negative sign (or magnitude 10.1 J clearly stated as energy removed).

2(B)(e)**[3 marks]**

What is the average force of air resistance exerted on the pine cone during the fall?

ANSWER

$$F_{\text{avg}} \approx 0.63 \text{ N, upwards}$$

RULES YOU NEED

- Work = force × displacement, so $F = W / d$
- $1 \text{ J} = 1 \text{ N m}$

WORKING — STEP BY STEP**1 Use $W = F \times d$**

Air resistance did 10.14 J of work (in size) over the whole 16 m fall. The average force is the work divided by the distance:

$$F_{\text{avg}} = |W_{\text{air}}| / h$$

2 Put the numbers in

$$F_{\text{avg}} = 10.14 \text{ J} \div 16 \text{ m} = 0.634 \text{ N}$$

3 Give the direction

It opposes the motion, so it points **upwards**.

4 Sense check

The cone's weight is $0.14 \times 9.81 = 1.37 \text{ N}$. The air resistance (0.63 N) is less than the weight, so the cone still speeds up as it falls, which matches it landing at 13 m/s.

Marks	Awarded for
1	Uses $W = F \times d$ with the work from (d).
1	Correct magnitude $\approx 0.63 \text{ N}$ (own-figure accepted).
1	Direction stated: upwards / opposite to the motion.

Question 2(B) Total: 10 Marks

Question 3 · Option A

Specific heat capacity, radiation, calorimetry — 10 marks

Given: Core body temperature lowered from 37 °C to 34 °C over 2 hours; $c_{\text{body}} = 3.5 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$; 65 kg person; 200 L bath; $c_{\text{water}} = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$.

3(A)(a)

[1 mark]

Define the quantity “specific heat capacity” and state its SI units.

ANSWER

c = heat per unit mass per kelvin; SI unit $\text{J kg}^{-1} \text{K}^{-1}$

RULES YOU NEED

- $Q = m c \Delta T$

WORKING — STEP BY STEP

1 Definition

The specific heat capacity c is the heat needed per unit mass to raise the temperature of a substance by one degree (1 K).

2 Units from the formula

Rearrange $Q = mc\Delta T$ to get $c = Q / (m \Delta T)$. Q is in J, m in kg, ΔT in K:

$$\text{units of } c = \text{J} \div (\text{kg} \times \text{K}) = \text{J kg}^{-1} \text{K}^{-1}$$

Marks

Awarded for

1

Definition **and** unit $\text{J kg}^{-1} \text{K}^{-1}$.

3(A)(b)**[2 marks]**

What is the fractional change in the power radiated by a person when their skin temperature drops from 37 °C to 36 °C?

ANSWER

Fractional change ≈ -0.013 : the radiated power falls by about 1.3%

RULES YOU NEED

- Stefan-Boltzmann: $P = A \epsilon \sigma T^4$
- T must be in kelvin: $T_K = T_{°C} + 273.15$
- Fractional change = (new – old) / old

WORKING — STEP BY STEP**1 Only T changes**

A (area), ϵ (emissivity) and σ (a constant) stay the same, so $P \propto T^4$.

2 Convert both temperatures to kelvin

$$T_1 = 37 + 273.15 = 310.15 \text{ K}$$

$$T_2 = 36 + 273.15 = 309.15 \text{ K}$$

3 Find the ratio of the powers

Because $P \propto T^4$, the constants cancel in the ratio:

$$P_2 / P_1 = (T_2 / T_1)^4 = (309.15 / 310.15)^4 = (0.996776)^4 = 0.98717$$

4 Fractional change

$$(P_2 - P_1) / P_1 = P_2/P_1 - 1 = 0.98717 - 1 = -0.0128$$

So the radiated power falls by about 1.3%.

EXAM TECHNIQUE

Using °C in the T^4 ratio, $(36/37)^4 = 0.896$, gives a 10% drop. That is wrong and loses the method mark. The T^4 law only works in kelvin.

Marks	Awarded for
1	$P \propto T^4$ with temperatures converted to kelvin.
1	Correct fractional change ≈ -0.013 (1.3% decrease).

3(A)(c)**[2 marks]**

What total amount of heat must be removed to drop the whole body temperature of a typical 65 kg person by 1.0 °C?

ANSWER

$$Q \approx 2.3 \times 10^5 \text{ J (227.5 kJ)}$$

RULES YOU NEED

- $Q = m c \Delta T$
- A temperature **difference** of 1 °C = 1 K

WORKING — STEP BY STEP**1 Write down the values**

$$m = 65 \text{ kg}, c = 3.5 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}, \Delta T = 1.0 \text{ °C} = 1.0 \text{ K}.$$

2 Apply $Q = mc\Delta T$

$$Q = 65 \times 3500 \times 1.0 = 227\,500 \text{ J}$$

3 Write it neatly

$$Q = 2.28 \times 10^5 \text{ J (228 kJ)}$$

Marks	Awarded for
1	$Q = mc\Delta T$ applied.
1	Correct value in J.

3(A)(d)**[1 mark]**

What is the average rate of heat loss, assuming the whole body's temperature changes by the same amount as the core temperature?

ANSWER

$$P \approx 95 \text{ W}$$

RULES YOU NEED

- $Q = m c \Delta T$
- Power = energy $\div$ time, $P = Q / t$
- 1 hour = 3600 s

WORKING — STEP BY STEP**1 Temperature change of the whole body**

The core goes from 37 °C to 34 °C, so $\Delta T = 3 \text{ K}$ for the whole body.

2 Total heat removed

$$Q = m c \Delta T = 65 \times 3500 \times 3 = 682\,500 \text{ J}$$

3 Time in seconds

$$t = 2 \text{ h} = 2 \times 3600 = 7200 \text{ s}$$

4 Rate of heat loss = power

$$P = Q / t = 682\,500 \div 7200 = 94.8 \text{ W}$$

Marks**Awarded for****1**Uses the full 3 K drop over 7200 s, giving $\approx 95 \text{ W}$.

3(A)(e)**[4 marks]**

The person is placed in a thermally isolated bath of 200 litres of water. If the water and body temperatures are the same at the end, what was the initial temperature of the water? ($c_{\text{water}} = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$)

ANSWER

Initial water temperature $\approx 33.2 \text{ }^{\circ}\text{C}$

RULES YOU NEED

- Thermally isolated: heat lost by one = heat gained by the other ($Q_{\text{gained}} = -Q_{\text{lost}}$)
- $Q = m c \Delta T$
- $1 \text{ L} = 0.001 \text{ m}^3$, $\rho_{\text{water}} = 1000 \text{ kg/m}^3$

WORKING — STEP BY STEP**1 Where does the heat go?**

The bath is isolated, so no heat escapes. Every joule the body loses goes into the water.

2 Heat lost by the body

The body cools from $37 \text{ }^{\circ}\text{C}$ to $34 \text{ }^{\circ}\text{C}$. From part (d):

$$Q_{\text{lost}} = 6.825 \times 10^5 \text{ J}$$

3 Mass of the water

$$V = 200 \text{ L} = 200 \times 0.001 \text{ m}^3 = 0.200 \text{ m}^3$$

$$m_{\text{w}} = \rho V = 1000 \times 0.200 = 200 \text{ kg}$$

4 How much does the water warm up?

The water gains the same $6.825 \times 10^5 \text{ J}$. Rearrange $Q = mc\Delta T$ for ΔT :

$$\Delta T_{\text{water}} = Q / (m_{\text{w}} c_{\text{w}}) = 682\,500 \div (200 \times 4200) = 682\,500 \div 840\,000 = 0.81 \text{ K}$$

5 Work back to the starting temperature

The water ends at the body's final temperature, $34 \text{ }^{\circ}\text{C}$, after warming by $0.81 \text{ }^{\circ}\text{C}$:

$$T_{\text{w,initial}} = 34 - 0.81 = 33.2 \text{ }^{\circ}\text{C}$$

EXAM TECHNIQUE

Sanity-check the direction. The water gains heat, so it must start *colder* than 34 °C. An answer above 34 °C means a sign error.

Marks	Awarded for
1	Heat balance: heat lost by body = heat gained by water.
1	Correct body heat (6.8×10^5 J) and water mass (200 kg).
1	Correct temperature rise of the water (≈ 0.81 K).
1	Correct initial temperature ≈ 33.2 °C.

Question 3(A) Total: 10 Marks

Question 3 · Option B

Pascal's principle, hydrostatic pressure, the saline drip, hydraulic lift — 10 marks

Given: Gauge pressure in a vein $2.4 \times 10^3 \text{ Pa}$; saline density $\rho = 1 \text{ g cm}^{-3}$; bag surface area 0.04 m^2 ; bed piston 0.12 m^2 , foot piston 0.03 m^2 .

3(B)(a)

[2 marks]

State Pascal's Principle.

ANSWER

Pressure applied to an enclosed fluid is transmitted undiminished to every part of the fluid and to the container walls.

RULES YOU NEED

- Pressure = force ÷ area ($\text{Pa} = \text{N m}^{-2}$)

WORKING — STEP BY STEP

1 The statement

The pressure applied at one point in an enclosed fluid is transmitted **undiminished** to every part of the fluid and to the walls of the container.

2 What it means in practice

Push on one piston of a sealed, liquid-filled system and the pressure rises by the same amount everywhere in the liquid. This is how a hydraulic lift works (part (e)).

Marks	Awarded for
1	Pressure applied to an enclosed fluid is transmitted to every part of it.
1	“Undiminished” (equal everywhere), including to the container walls.

3(B)(b)**[2 marks]**

Calculate the minimum height the drip must be above the insertion point for the saline solution to enter the vein.

ANSWER

$$h_{\min} \approx 0.24 \text{ m (24 cm)}$$

RULES YOU NEED

- Hydrostatic pressure: $P = \rho g h$
- $1 \text{ g cm}^{-3} = 1000 \text{ kg m}^{-3}$

WORKING — STEP BY STEP**1 What is the condition?**

The saline only flows in if the pressure it creates at the needle is at least the pressure in the vein, 2400 Pa.

2 Convert the density to SI

$1 \text{ g} = 10^{-3} \text{ kg}$ and $1 \text{ cm}^3 = 10^{-6} \text{ m}^3$:

$$\rho = 1 \text{ g cm}^{-3} = 10^{-3} \text{ kg} \div 10^{-6} \text{ m}^3 = 1000 \text{ kg m}^{-3}$$

3 Rearrange $P = \rho g h$ for h

$$h = P / (\rho g)$$

4 Put the numbers in

$$h = 2400 \div (1000 \times 9.81) = 2400 \div 9810 = 0.245 \text{ m}$$

Marks	Awarded for
1	$\rho g h = P_{\text{vein}}$ with ρ in kg m^{-3} .
1	Correct height $\approx 0.24 \text{ m}$.

3(B)(c)**[2 marks]**

The drip is suspended 0.2 m above the insertion point. What is the total pressure exerted by the saline at the insertion point?

ANSWER

$$P \approx 1.96 \times 10^3 \text{ Pa (gauge)}$$

RULES YOU NEED

- $P = \rho g h$ (pressure from a column of liquid)
- Gauge pressure = pressure above atmospheric

WORKING — STEP BY STEP**1 Pressure from 0.2 m of saline**

$$P = \rho g h = 1000 \times 9.81 \times 0.2 = 1962 \text{ Pa}$$

2 Gauge or absolute?

The vein pressure (2400 Pa) is a gauge pressure, so give the saline pressure as a gauge pressure too, to compare like with like.

3 Compare with the vein

1962 Pa is less than 2400 Pa, so at this height the saline will **not** flow in by itself.

EXAM TECHNIQUE

If you add atmospheric pressure, give it separately: $P_{\text{abs}} = 1.013 \times 10^5 + 1962 \approx 1.03 \times 10^5 \text{ Pa}$.
Either is accepted if clearly labelled.

Marks	Awarded for
1	$P = \rho g h$ with $h = 0.2 \text{ m}$.
1	Correct value $\approx 1960 \text{ Pa}$.

3(B)(d)**[2 marks]**

The additional pressure could be provided by squeezing the bag. What force must be applied? (Force spread over the whole 0.04 m^2 bag.)

ANSWER

$$F \approx 17.5 \text{ N}$$

RULES YOU NEED

- Pressure = force $\div$ area, so $F = P \times A$

WORKING — STEP BY STEP**1 How much extra pressure is needed?**

$$\Delta P = P_{\text{vein}} - P_{\text{saline}} = 2400 - 1962 = 438 \text{ Pa}$$

2 Turn pressure into force

$$F = \Delta P \times A = 438 \text{ Pa} \times 0.04 \text{ m}^2 = 17.5 \text{ N}$$

3 Sense check

17.5 N is roughly the weight of a 1.8 kg bag of sugar: a firm squeeze, which is realistic.

Marks	Awarded for
1	Additional pressure $\Delta P = 438 \text{ Pa}$.
1	$F = \Delta P \times A \approx 17.5 \text{ N}$ (own-figure from (c) accepted).

3(B)(e)**[2 marks]**

Alternatively, the bed could be lowered with a hydraulic lift (bed piston 0.12 m^2 , foot piston 0.03 m^2). How much does the foot piston's height increase to drop the bed by the required height?

ANSWER

The foot piston rises $\approx 0.18 \text{ m}$ (18 cm)

RULES YOU NEED

- Liquids are incompressible: volume out of one cylinder = volume into the other
- Volume of a piston stroke = area $\times$ height

WORKING — STEP BY STEP**1 How far must the bed drop?**

Lowering the patient raises the drip higher above the needle. It must go from 0.20 m to the 0.245 m found in (b):

$$\Delta h_{\text{bed}} = 0.2446 - 0.20 = 0.0446 \text{ m}$$

2 Volume of fluid pushed out by the bed piston

$$V = A_{\text{bed}} \times \Delta h_{\text{bed}} = 0.12 \times 0.0446 = 5.35 \times 10^{-3} \text{ m}^3$$

3 That volume goes into the foot cylinder

The fluid can't be squashed, so the same volume raises the foot piston:

$$\Delta h_{\text{foot}} = V / A_{\text{foot}} = 5.35 \times 10^{-3} \div 0.03 = 0.178 \text{ m}$$

4 Check with the area ratio

The foot piston is 4 times smaller ($0.12/0.03 = 4$), so it moves 4 times as far: $4 \times 0.0446 = 0.178 \text{ m}$ ✓

Marks	Awarded for
1	Required drop of the bed $\approx 0.045 \text{ m}$, from (b) – 0.2 m .
1	Equal volumes, giving $\Delta h_{\text{foot}} \approx 0.18 \text{ m}$ (own-figure accepted).

Question 3(B) Total: 10 Marks

Question 4 · Option A

Electric field, charge density and energy of an ion crossing a neuron membrane — 10 marks

Given: Membrane thickness $0.12 \mu\text{m}$; potential difference outside to inside -70 mV ; inside and outside walls oppositely charged (+ outside, – inside).

4(A)(a)

[1 mark]

What is the SI unit for electric field strength?

ANSWER

V m^{-1} (equivalently N C^{-1})

RULES YOU NEED

- Force on a charge: $F = qE$
- Work per unit charge over distance d : $W/q = Fd/q = Ed = V$

WORKING — STEP BY STEP

1 Link E to voltage

Moving a charge q a distance d along a field E , the work per unit charge is $Fd/q = Ed$. That is the potential difference V , so $V = Ed$.

2 Rearrange for E

$$E = V / d \rightarrow \text{units: volts} \div \text{metres} = \text{V m}^{-1}$$

Equivalently, from $F = qE$: $E = F/q$, in N C^{-1} . The two units are the same thing.

Marks

Awarded for

1

V m^{-1} (or N C^{-1}).

4(A)(b)**[2 marks]**

Calculate the electric field inside the cell membrane, in SI units.

ANSWER

$|E| \approx 5.8 \times 10^5 \text{ V m}^{-1}$, directed from outside to inside

RULES YOU NEED

- Uniform field between two charged layers: $V = Ed$, so $E = V/d$
- $1 \text{ mV} = 10^{-3} \text{ V}$; $1 \mu\text{m} = 10^{-6} \text{ m}$

WORKING — STEP BY STEP**1 Treat the membrane like two charged plates**

The outside wall is positive and the inside wall negative, like a capacitor. Between them the field is uniform, so $V = Ed$.

2 Convert to SI units

$$V = 70 \text{ mV} = 70 \times 10^{-3} \text{ V} = 0.070 \text{ V}$$

$$d = 0.12 \mu\text{m} = 0.12 \times 10^{-6} \text{ m} = 1.2 \times 10^{-7} \text{ m}$$

3 Calculate E

$$E = V / d = 0.070 \div 1.2 \times 10^{-7} = 5.83 \times 10^5 \text{ V m}^{-1}$$

4 Direction

Field lines go from positive to negative: from the outside wall towards the inside wall, i.e. **into** the cell.

Marks	Awarded for
1	$E = V/d$ with μm and mV converted to SI.
1	Correct magnitude with unit.

4(A)(c)**[1 mark]**

Calculate the magnitude of the charge density on the inside of the membrane wall, in C/m².

ANSWER

$$\sigma \approx 5.2 \times 10^{-6} \text{ C m}^{-2}$$

RULES YOU NEED

- Field between charged plates: $E = Q / (\epsilon_0 A)$
- Charge per unit area $\sigma = Q/A$
- $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ J}^{-1} \text{ m}^{-1}$

WORKING — STEP BY STEP**1 Rewrite the plate formula in terms of σ**

Q/A is the charge per unit area, σ . So $E = \sigma/\epsilon_0$, which rearranges to:

$$\sigma = \epsilon_0 E$$

2 Put the numbers in

$$\sigma = 8.854 \times 10^{-12} \times 5.83 \times 10^5 = 5.16 \times 10^{-6} \text{ C m}^{-2}$$

Marks**Awarded for****1**

$\sigma = \epsilon_0 E \approx 5.2 \times 10^{-6} \text{ C m}^{-2}$ (own-figure from (b) accepted).

4(A)(d)**[3 marks]**

A positive sodium ion ($m = 3.81 \times 10^{-26}$ kg, $q = 1.6 \times 10^{-19}$ C) starts from rest on the outside wall. Calculate the change in its potential energy once it reaches the inside wall.

ANSWER

$$\Delta U = -1.1 \times 10^{-20} \text{ J (a decrease of } 1.12 \times 10^{-20} \text{ J)}$$

RULES YOU NEED

- Work done moving a charge through a potential difference: $W = Vq$
- Change in electric PE: $\Delta U = q \Delta V$
- Charges move freely from higher to lower electric PE (like a falling object)

WORKING — STEP BY STEP**1 Which side is at higher potential?**

The outside is positive and the inside negative, so the inside is 70 mV **lower** in potential than the outside.

2 Potential change for the ion's trip

Going from outside to inside, the potential drops by 70 mV:

$$\Delta V = V_{\text{inside}} - V_{\text{outside}} = -0.070 \text{ V}$$

3 Change in potential energy

$$\Delta U = q \Delta V = 1.6 \times 10^{-19} \text{ C} \times (-0.070 \text{ V}) = -1.12 \times 10^{-20} \text{ J}$$

4 Interpret the sign

Negative means the PE **decreases**: a positive ion moving to lower potential, just like a ball rolling downhill.

Marks	Awarded for
1	$\Delta U = q\Delta V$ (or $W = Vq$).
1	Correct magnitude 1.12×10^{-20} J.
1	Correct sign: potential energy decreases (with reason: positive ion moving to lower potential).

4(A)(e)**[3 marks]**

What is the velocity of the sodium ion just before it hits the inside wall, if no other force acts?

ANSWER

$v \approx 7.7 \times 10^2 \text{ m s}^{-1}$ (767 m/s), directed into the cell

RULES YOU NEED

- $E_{\text{kin}} = \frac{1}{2} m v^2$
- Only one force acting $\Rightarrow$ energy conserved: PE lost = KE gained

WORKING — STEP BY STEP**1 Where does the lost PE go?**

No other force acts, so all $1.12 \times 10^{-20} \text{ J}$ of lost potential energy becomes kinetic energy. The ion started at rest, so its KE starts at 0.

2 Set KE equal to the PE lost

$$\frac{1}{2} m v^2 = 1.12 \times 10^{-20} \text{ J}$$

3 Rearrange for v

$$v = \sqrt{(2 \times 1.12 \times 10^{-20} \div 3.81 \times 10^{-26})}$$

4 Calculate step by step

$$2 \times 1.12 \times 10^{-20} = 2.24 \times 10^{-20}$$

$$2.24 \times 10^{-20} \div 3.81 \times 10^{-26} = 5.88 \times 10^5$$

$$v = \sqrt{(5.88 \times 10^5)} = 767 \text{ m/s}$$

Marks	Awarded for
1	Energy conservation: loss in PE = gain in KE.
1	Rearranged correctly for v.
1	Correct $v \approx 767 \text{ m s}^{-1}$ (own-figure from (d) accepted).

Question 4(A) Total: 10 Marks

Study-Uni

Question 4 · Option B

Ohm's law, a two-resistor circuit, electrical power, current as charge flow — 10 marks

Given: 12 V supply; $R_1 = 5\ \Omega$; current at point a is 2 A; R_1 and R_2 are in one loop, so the same 2 A flows through both.

4(B)(a)

[2 marks]

State and explain Ohm's Law. Give an example of when Ohm's law is applicable.

ANSWER

**$V = IR$: current is proportional to voltage for a fixed resistance.
Applies e.g. to metal resistors / heating elements.**

RULES YOU NEED

- $V = I R$
- For a given V : $I \propto 1/R$
- $R = \rho l/A$ (resistance depends on the material and its shape)

WORKING — STEP BY STEP

1 State the law

$$V = I R$$

The current through a conductor is proportional to the voltage across it. The constant of proportionality is the resistance R .

2 Explain what it means

R measures how strongly the material opposes the flow of charge. If R stays constant, doubling V doubles I . For a fixed V , a bigger R means a smaller current ($I \propto 1/R$).

3 Give an example

A metal heating element, e.g. the lecture's 2 kW kettle on 220 V: $R = V^2/P = 220^2 \div 2000 = 24\ \Omega$, and the current is $I = V/R = 220 \div 24 \approx 9\text{ A}$.

Note on the lecture material: The lecture notes use $V = IR$ throughout but don't give a formal written statement of "Ohm's law" or its conditions (e.g. constant temperature). The answer is built only from how the lecturer uses $V = IR$ and $I \propto 1/R$.

Marks	Awarded for
1	$V = IR$ stated, with $I \propto V$ at constant R explained.
1	Valid example of an ohmic situation (resistor, heating element, skin resistance...).

4(B)(b)**[2 marks]**Calculate the resistance of the second resistor, R_2 .**ANSWER**

$$R_2 = 1 \, \Omega$$

RULES YOU NEED

- $V = IR$
- V = work done per unit charge: the battery's 12 V is shared out around the loop

WORKING — STEP BY STEP**1 Same current everywhere in one loop**

There is only one path, so the 2 A measured at a flows through R_1 and R_2 .

2 Voltage across R_1

$$V_1 = IR_1 = 2 \, \text{A} \times 5 \, \Omega = 10 \, \text{V}$$

3 Voltage left for R_2

The battery gives each coulomb 12 J. R_1 uses 10 J of that, so R_2 gets the rest:

$$V_2 = 12 - 10 = 2 \, \text{V}$$

4 Resistance of R_2

$$R_2 = V_2 / I = 2 \, \text{V} \div 2 \, \text{A} = 1 \, \Omega$$

5 Check

Whole circuit: $R_{\text{total}} = 12 \, \text{V} \div 2 \, \text{A} = 6 \, \Omega$, and $5 \, \Omega + 1 \, \Omega = 6 \, \Omega \checkmark$

Marks	Awarded for
1	Correct method: $V_1 = IR_1$ and the remaining voltage across R_2 (or $R_{\text{total}} = V/I = 6 \, \Omega$).
1	$R_2 = 1 \, \Omega$.

4(B)(c)**[2 marks]**

Calculate the magnitude of the voltage across each of the resistors.

ANSWER

$$V_1 = 10 \text{ V}; V_2 = 2 \text{ V}$$

RULES YOU NEED

- $V = IR$

WORKING — STEP BY STEP**1 Across R_1**

$$V_1 = 2 \text{ A} \times 5 \Omega = 10 \text{ V}$$

2 Across R_2

$$V_2 = 2 \text{ A} \times 1 \Omega = 2 \text{ V}$$

3 Check they add up to the supply

$$10 \text{ V} + 2 \text{ V} = 12 \text{ V} \checkmark$$

Marks	Awarded for
1	$V_1 = 10 \text{ V}$.
1	$V_2 = 2 \text{ V}$ (own-figure from (b) accepted).

4(B)(d)**[2 marks]**

Calculate the power used by each resistor.

ANSWER

$P_1 = 20 \text{ W}; P_2 = 4 \text{ W}$

RULES YOU NEED

- $P = I V$
- $P = I^2 R = V^2 / R$ (using $V = IR$)

WORKING — STEP BY STEP**1 Power in R_1**

$$P_1 = I^2 R_1 = 2^2 \times 5 = 4 \times 5 = 20 \text{ W}$$

Check with $P = IV$: $2 \times 10 = 20 \text{ W} \checkmark$ **2 Power in R_2**

$$P_2 = I^2 R_2 = 2^2 \times 1 = 4 \text{ W}$$

Check: $2 \times 2 = 4 \text{ W} \checkmark$ **3 Check the total against the battery**

Battery power: $P = IV = 2 \times 12 = 24 \text{ W}$, and $20 + 4 = 24 \text{ W} \checkmark$. In each resistor this energy is turned into heat.

Marks	Awarded for
1	$P_1 = 20 \text{ W}.$
1	$P_2 = 4 \text{ W}.$

4(B)(e)**[2 marks]**

Calculate the number of electrons that flow past the point marked a in 1 second.

ANSWER

$N \approx 1.25 \times 10^{19}$ electrons per second

RULES YOU NEED

- $I = Q / t$, so $Q = I t$
- Charge of one electron $e = 1.602 \times 10^{-19} \text{ C}$
- Number of electrons = Q / e

WORKING — STEP BY STEP**1 Charge passing in 1 s**

A current of 2 A means 2 coulombs pass every second:

$$Q = I t = 2 \text{ A} \times 1 \text{ s} = 2 \text{ C}$$

2 Divide by the charge on one electron

$$N = Q / e = 2 \div 1.602 \times 10^{-19} = 1.25 \times 10^{19}$$

Marks	Awarded for
1	$Q = I t = 2 \text{ C}.$
1	$N = Q/e \approx 1.25 \times 10^{19}.$

Question 4(B) Total: 10 Marks

Question 5 · Option A

Earth's magnetic poles, Lorentz force, circular motion of a charge — 10 marks

Given: $q = +5 \text{ mC}$, $m = 1 \times 10^{-10} \text{ kg}$, $v = 3 \times 10^4 \text{ m s}^{-1}$ left $\rightarrow$ right, $B = 5 \text{ } \mu\text{T}$ into the page.

5(A)(a)

[2 marks]

The magnetic north pole on a compass points towards the geographic north pole. Explain why this means the Earth's magnetic south pole must lie in the direction of the geographic north pole.

ANSWER

Unlike poles attract. The compass's N pole is pulled north, so the pole there must be a magnetic S pole.

RULES YOU NEED

- A compass is a small dipole magnet with a N and a S pole.
- Similar poles repel; unlike poles attract.

WORKING — STEP BY STEP

1 What pulls the compass needle?

The needle's north pole is pulled towards the geographic north.

2 Which kind of pole attracts a north pole?

Like poles repel and unlike poles attract. Only a **south** magnetic pole attracts a north pole.

3 Conclusion

So the magnetic pole near the geographic north must be a magnetic **south** pole (and the magnetic north pole is near the geographic south). The lecture diagram of Earth's dipole shows exactly this.

Marks	Awarded for
1	Unlike poles attract (compass is a magnetic dipole).
1	So the pole that attracts the compass's N end must be a magnetic S pole, at geographic north.

5(A)(b)**[2 marks]**

Draw a copy of the diagram and include the direction of the Lorentz force that the particle feels.

ANSWER

Lorentz force points up the page (perpendicular to v , in the plane of the page)

RULES YOU NEED

- The force on a moving charge is perpendicular to **both** v and B .
- Left Hand Rule: First finger = Field, seCond finger = Current (direction of + charge's motion), thuMb = Motion/force.

WORKING — STEP BY STEP**1 Set up your hand**

Point your first finger into the page (B). Point your second finger to the right (the positive charge moves right).

2 Read the thumb

Your thumb points **up the page**. That is the direction of the force.

3 Draw it

Copy the grid of $\otimes$ symbols, the particle and its velocity arrow, then add a force arrow pointing up from the particle.

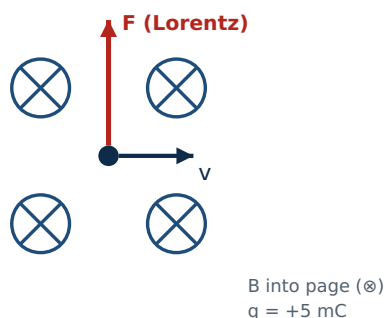


Fig. 1 — Positive charge moving right through B into the page: the Lorentz force points up the page.

4 Check against the lecture example

In the notes, a $+Q$ moving up the page in a field into the page is pushed to the left. Turn that picture a quarter-turn clockwise and you get this case, with the force pointing up ✓

Marks	Awarded for
1	Diagram reproduced with B into page and v to the right.
1	Force correctly drawn up the page, perpendicular to v.

5(A)(c)**[2 marks]**

Calculate the magnitude of the Lorentz force experienced by the particle.

ANSWER

$$\mathbf{F = 7.5 \times 10^{-4} \text{ N}}$$

RULES YOU NEED

- $F = B Q v$ (for v perpendicular to B)
- $1 \text{ mC} = 10^{-3} \text{ C}$; $1 \mu\text{T} = 10^{-6} \text{ T}$

WORKING — STEP BY STEP**1 Convert to SI units**

$$Q = 5 \text{ mC} = 5 \times 10^{-3} \text{ C}$$

$$B = 5 \mu\text{T} = 5 \times 10^{-6} \text{ T}$$

2 Apply $F = BQv$

$$F = 5 \times 10^{-6} \times 5 \times 10^{-3} \times 3 \times 10^4$$

3 Multiply the numbers and the powers of ten separately

$$5 \times 5 \times 3 = 75, \text{ and } 10^{-6} \times 10^{-3} \times 10^4 = 10^{-5}:$$

$$F = 75 \times 10^{-5} \text{ N} = 7.5 \times 10^{-4} \text{ N}$$

Marks	Awarded for
1	$F = BQv$ with μT and mC converted.
1	Correct value $7.5 \times 10^{-4} \text{ N}$.

5(A)(d)**[3 marks]**

The particle undergoes uniform circular motion in the field. Calculate the radius of this motion.

ANSWER

$$r = 120 \text{ m}$$

RULES YOU NEED

- Centripetal force needed for circular motion: $F = mv^2/r$
- Here the Lorentz force provides it: $F = BQv$
- So $r = mv / (QB)$

WORKING — STEP BY STEP**1 What keeps it moving in a circle?**

The Lorentz force always points sideways to the motion, so it acts as the centripetal force.

2 Set the two forces equal

$$m v^2 / r = B Q v$$

3 Rearrange for r

Divide both sides by v , then solve for r :

$$r = m v / (Q B)$$

4 Put the numbers in

$$\text{top: } m v = 1 \times 10^{-10} \times 3 \times 10^4 = 3 \times 10^{-6}$$

$$\text{bottom: } Q B = 5 \times 10^{-3} \times 5 \times 10^{-6} = 2.5 \times 10^{-8}$$

$$r = 3 \times 10^{-6} \div 2.5 \times 10^{-8} = 120 \text{ m}$$

Marks	Awarded for
1	Lorentz force set equal to centripetal force.
1	Rearranged to $r = mv/(QB)$.
1	$r = 120 \text{ m}$.

5(A)(e)**[1 mark]**

If the particle were replaced with one of opposite charge moving in the opposite direction, would the direction of the Lorentz force stay the same or change? Explain briefly.

ANSWER**It stays the same: the two reversals cancel****RULES YOU NEED**

- Current direction = direction of positive charge flow; negative charges moving one way = current the other way.

WORKING — STEP BY STEP**1 Effect of flipping the charge**

Reversing the sign of the charge reverses the force.

2 Effect of flipping the velocity

Reversing the direction of motion also reverses the force.

3 Both at once

Two reversals cancel. Equivalently, a negative charge moving left is the same current as a positive charge moving right, so the force is **the same**.

Marks**Awarded for****1**

"Same", with the double-reversal / equivalent-current reasoning.

Question 5(A) Total: 10 Marks

Question 5 · Option B

Magnetic field of a wire, resistivity, force between parallel currents — 10 marks

Given: Two parallel 5 m wires, 2 m apart, each on a 12 V supply, each carrying 2.5 A in the same direction; wire radius 2 mm.

5(B)(a)

[2 marks]

Draw a diagram of this setup, labelling the direction of the current in each wire and the direction of the magnetic field on both sides of each wire.

ANSWER

Currents drawn parallel and in the same direction; B out of page on the left and into page on the right of each wire

RULES YOU NEED

- Right-hand rule: grip the wire with your right hand, thumb along I; your fingers curl in the direction of B.

WORKING — STEP BY STEP

1 Draw the wires and currents

Two parallel lines 2 m apart, both with current arrows pointing the same way (e.g. both up the page).

2 Use the right-hand rule on one wire

Thumb up (along I). Your fingers come out of the page on the left of the wire and go into the page on the right.

3 Mark the field

Draw ⊙ (out of page) on the left of each wire and ⊗ (into page) on the right. The same pattern applies to both wires.

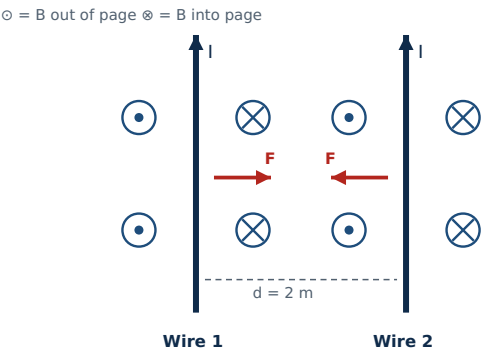


Fig. 2 — Both currents up the page: B out of the page (⊙) on the left of each wire, into the page (⊗) on the right. The wires attract.

Marks	Awarded for
1	Both currents shown in the same direction.
1	B directions correct on both sides of both wires.

5(B)(b)**[3 marks]**

Calculate the resistivity of the material the wires are made of.

ANSWER

$$\rho \approx 1.2 \times 10^{-5} \Omega \text{ m}$$

RULES YOU NEED

- $V = I R$, so $R = V / I$
- Resistance and resistivity: $R = \rho \ell / A$, so $\rho = R A / \ell$
- Area of a circle $A = \pi r^2$

WORKING — STEP BY STEP**1 Resistance of each wire**

$$R = V / I = 12 \text{ V} \div 2.5 \text{ A} = 4.8 \Omega$$

2 Cross-sectional area of the wireRadius 2 mm = $2 \times 10^{-3} \text{ m}$:

$$A = \pi r^2 = \pi \times (2 \times 10^{-3})^2 = \pi \times 4 \times 10^{-6} = 1.257 \times 10^{-5} \text{ m}^2$$

3 Rearrange $R = \rho \ell / A$ for ρ

$$\rho = R A / \ell$$

4 Put the numbers in ($\ell = 5 \text{ m}$)

$$\rho = 4.8 \times 1.257 \times 10^{-5} \div 5 = 1.21 \times 10^{-5} \Omega \text{ m}$$

Marks	Awarded for
1	$R = V/I = 4.8 \Omega$.
1	$A = \pi r^2$ in m^2 .
1	$\rho = RA/\ell \approx 1.2 \times 10^{-5} \Omega \text{ m}$.

5(B)(c)**[2 marks]**

Calculate the magnitude of the magnetic field B at wire 2 due to the current in wire 1.

ANSWER

$$B = 2.5 \times 10^{-7} \text{ T}$$

RULES YOU NEED

- Field at distance d from a long wire: $B = (\mu_0/2\pi) \cdot I/d$
- $\mu_0 = 4\pi \times 10^{-7} \text{ N A}^{-2}$, so $\mu_0/2\pi = 2 \times 10^{-7}$

WORKING — STEP BY STEP**1 Simplify the constant first**

$$\mu_0 / 2\pi = 4\pi \times 10^{-7} \div 2\pi = 2 \times 10^{-7} \text{ N A}^{-2}$$

2 Apply the formula with $I = 2.5 \text{ A}$, $d = 2 \text{ m}$

$$B = 2 \times 10^{-7} \times 2.5 \div 2 = 2 \times 10^{-7} \times 1.25 = 2.5 \times 10^{-7} \text{ T}$$

Marks	Awarded for
1	Correct formula $B = \mu_0 I / (2\pi d)$.
1	$B = 2.5 \times 10^{-7} \text{ T}$.

5(B)(d)**[2 marks]**

Calculate the magnitude of the force exerted on each wire.

ANSWER

$F \approx 3.1 \times 10^{-6} \text{ N}$ on each wire

RULES YOU NEED

- Force on a current in a field ($I \perp B$): $F = B I \ell$
- Newton's third law: equal and opposite forces on the two wires

WORKING — STEP BY STEP**1 Wire 2 sits in wire 1's field**

Wire 2 carries 2.5 A, perpendicular to the field from (c), along its whole 5 m length.

2 Apply $F = BI\ell$

$$F = 2.5 \times 10^{-7} \text{ T} \times 2.5 \text{ A} \times 5 \text{ m} = 3.1 \times 10^{-6} \text{ N}$$

3 The other wire

By Newton's third law, wire 1 feels the same size of force from wire 2.

Marks	Awarded for
1	$F = BI\ell$ (or F/ℓ formula) with $\ell = 5 \text{ m}$.
1	$F \approx 3.1 \times 10^{-6} \text{ N}$ (own-figure from (c) accepted).

5(B)(e)**[1 mark]**

Do the wires move towards each other or away from each other due to this force?

ANSWER**Towards each other (attractive)****RULES YOU NEED**

- Currents in the **same** direction attract; opposite directions repel.

WORKING — STEP BY STEP**1 Check the current directions**

Both currents flow the same way.

2 Apply the rule

Same direction → the Lorentz force on each wire points towards the other → they **attract**.

Marks	Awarded for
1	Towards each other.

Question 5(B) Total: 10 Marks

Question 6 · Option A

Transverse vs longitudinal waves, $v = f\lambda$, two-source interference — 10 marks

Given: Speakers $D = 2.01\text{ m}$ apart, in phase, $\lambda = 0.9\text{ m}$; microphone $d_1 = 1.8\text{ m}$ directly in front of speaker 1; speed of sound 343 m s^{-1} .

6(A)(a)

[2 marks]

What is the difference between transverse waves and longitudinal waves? Give an example for each type.

ANSWER

Transverse: oscillation $\perp$ to propagation (e.g. light, water waves). Longitudinal: oscillation $\parallel$ to propagation (e.g. sound).

RULES YOU NEED

- A wave transfers energy without net motion of the medium.

WORKING — STEP BY STEP

1 Transverse waves

The medium moves **perpendicular** (at right angles) to the direction the wave travels. Example: light, water waves, a wave on a rope.

2 Longitudinal waves

The medium vibrates **parallel** to the direction the wave travels, making compressions (squashed) and rarefactions (stretched). Example: sound in air.

Marks	Awarded for
1	Correct distinction ($\perp$ vs $\parallel$ to the direction of propagation).
1	A correct example of each.

6(A)(b)**[1 mark]**

What is the frequency of the sound wave?

ANSWER

$f \approx 381 \text{ Hz}$

RULES YOU NEED

- $v = f \lambda$, so $f = v / \lambda$

WORKING — STEP BY STEP**1 Rearrange $v = f\lambda$ for f**

$$f = v / \lambda$$

2 Put the numbers in

$$f = 343 \text{ m/s} \div 0.9 \text{ m} = 381 \text{ Hz}$$

Marks**Awarded for****1** $f \approx 381 \text{ Hz}.$

6(A)(c)**[2 marks]**Calculate the distance d_2 , in metres.**ANSWER**

$d_2 \approx 2.70 \text{ m (2.698 m)}$

RULES YOU NEED

- Pythagoras: $\text{hypotenuse}^2 = \text{side}_1^2 + \text{side}_2^2$

WORKING — STEP BY STEP**1 Spot the right angle**

The microphone is directly in front of speaker 1, so d_1 and D meet at 90° . d_2 is the long side (hypotenuse).

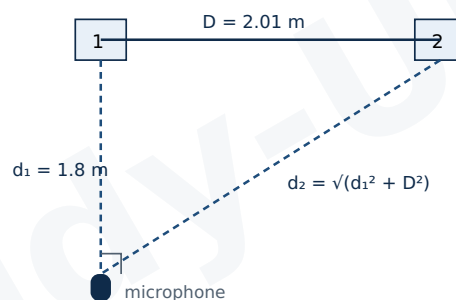


Fig. 3 — d_1 and D are at right angles, so d_2 is the hypotenuse.

2 Apply Pythagoras

$$d_2^2 = d_1^2 + D^2 = 1.8^2 + 2.01^2 = 3.24 + 4.0401 = 7.2801$$

3 Take the square root

$$d_2 = \sqrt{7.2801} = 2.698 \text{ m}$$

Marks	Awarded for
1	Pythagoras set up correctly.
1	$d_2 \approx 2.70 \text{ m}$.

6(A)(d)**[2 marks]**

What is the condition for constructive and destructive interference at the microphone in terms of $d_2 - d_1$?

ANSWER

Constructive: $d_2 - d_1 = n\lambda$. Destructive: $d_2 - d_1 = (n + \frac{1}{2})\lambda$.

RULES YOU NEED

- Waves in phase (crest on crest) add: constructive.
- Waves exactly out of phase (crest on trough) cancel: destructive.
- λ = distance from one crest to the next.

WORKING — STEP BY STEP**1 The speakers start in phase**

So what matters is how much further one wave travels: the path difference $d_2 - d_1$.

2 Constructive

If the extra distance is a whole number of wavelengths, the waves still line up crest-on-crest:

$$d_2 - d_1 = n\lambda \quad (n = 0, 1, 2, \dots)$$

3 Destructive

If the extra distance is an odd number of half-wavelengths, one arrives crest while the other arrives trough:

$$d_2 - d_1 = (n + \frac{1}{2})\lambda$$

Note on the lecture material: The notes show that in-phase waves add and exactly out-of-phase waves cancel, but don't write out the path-difference conditions $n\lambda$ and $(n + \frac{1}{2})\lambda$. These follow directly from that, together with the lecture definition of λ as the crest-to-crest distance.

Marks	Awarded for
1	Constructive: path difference = whole number of wavelengths.
1	Destructive: path difference = odd number of half-wavelengths.

6(A)(e)**[2 marks]**

Calculate whether the sound waves interfere constructively or destructively at the microphone.

ANSWER

Path difference $\approx 1\lambda \Rightarrow$ constructive interference

RULES YOU NEED

- Path difference $\div \lambda$: a whole number $\rightarrow$ constructive; a half-number (0.5, 1.5 ...) $\rightarrow$ destructive

WORKING — STEP BY STEP**1 Find the path difference**

$$d_2 - d_1 = 2.698 - 1.8 = 0.898 \text{ m}$$

2 Express it in wavelengths

$$0.898 \div 0.9 = 1.00 \text{ wavelengths}$$

3 Compare with the conditions

1.00 is a whole number ($n = 1$), so the waves arrive in phase: **constructive** interference.

Marks	Awarded for
1	Path difference $\approx 0.90 \text{ m} \approx 1.00\lambda$.
1	Concludes constructive (own-figure from (c) with consistent reasoning accepted).

6(A)(f)**[1 mark]**

Based on your previous answers, will the microphone pick up sounds or not?

ANSWER

Yes — a loud (reinforced) sound

RULES YOU NEED

- Constructive interference: amplitudes add (up to $2A$).

WORKING — STEP BY STEP**1 Use the result from (e)**

The interference is constructive, so the two waves add to a bigger amplitude at the microphone.

2 Conclusion

Yes: the microphone picks up a **loud** sound.

Marks	Awarded for
1	Yes, loud/reinforced sound, consistent with (e).

Question 6(A) Total: 10 Marks

Question 6 · Option B

Snell's law, critical angle, refraction through an aquarium wall — 10 marks

Given: $n_{\text{glass}} = 1.5$, $n_{\text{air}} = 1$, $n_{\text{water}} = 1.33$; ray hits the glass at $\theta_1 = 45^\circ$.

6(B)(a)

[3 marks]

State and explain Snell's law of refraction (including the definition of the refractive index).

ANSWER

$n_1 \sin \theta_1 = n_2 \sin \theta_2$ (angles from the normal, rays and normal coplanar); $n = c / v_{\text{medium}}$

RULES YOU NEED

- $n_1 \sin \theta_1 = n_2 \sin \theta_2$
- $n = c / v$ (speed of light in the medium)
- Angles are measured from the normal.

WORKING — STEP BY STEP

1 The law

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

θ_1 is the angle of incidence and θ_2 the angle of refraction, both measured from the **normal** (the line at 90° to the surface). The incident ray, refracted ray and normal all lie in the same plane.

2 The refractive index

$$n \equiv (\text{speed of light in vacuum, } c) \div (\text{speed of light in the medium})$$

Light is slower in denser materials, so n is bigger: vacuum = 1, air ≈ 1 , water ≈ 1.33 , glass ≈ 1.5 .

3 What it means

Going into a medium with a bigger n , the ray bends **towards** the normal. Going into a smaller n , it bends **away** from the normal.

Marks	Awarded for
1	Snell's law stated with angles defined from the normal.
1	Refractive index defined as $n = c/v$.
1	Explanation: bending towards/away from the normal, or rays and normal in the same plane.

6(B)(b)**[1 mark]**

Explain the concept of the “critical” angle in the context of Snell's law.

ANSWER

θ_{crit} = incidence angle (from higher-n to lower-n medium) giving 90° refraction; above it, total internal reflection

RULES YOU NEED

- From higher n to lower n , the ray bends away from the normal.

WORKING — STEP BY STEP**1 Going from dense to less dense**

When light passes from a higher n to a lower n (e.g. glass → air), it bends away from the normal, so the refracted angle is bigger than the incident angle.

2 The critical angle

As the incident angle grows, the refracted angle reaches 90°: the ray skims along the surface. That incident angle is the **critical angle** θ_{crit} .

3 Beyond it

For any bigger angle, no light gets out: it is all reflected back inside. This is **total internal reflection**, used in optical fibres.

Marks	Awarded for
1	Defined via 90° refraction angle when going from higher to lower n (TIR above it).

6(B)(c)**[1 mark]**

What is the general formula for the critical angle?

ANSWER

$$\theta_{\text{crit}} = \sin^{-1}(n_2/n_1)$$

RULES YOU NEED

- Snell's law with $\theta_2 = 90^\circ$
- $\sin 90^\circ = 1$

WORKING — STEP BY STEP**1 Put $\theta_2 = 90^\circ$ into Snell's law**

$$n_1 \sin \theta_{\text{crit}} = n_2 \sin 90^\circ = n_2$$

2 Rearrange

$$\theta_{\text{crit}} = \sin^{-1}(n_2 / n_1), \text{ with } n_1 > n_2$$

Marks**Awarded for****1**

$$\theta_{\text{crit}} = \sin^{-1}(n_2/n_1).$$

6(B)(d)**[2 marks]**

What is the critical angle for the transition from glass ($n = 1.5$) to air ($n = 1$)?

ANSWER

$$\theta_{\text{crit}} \approx 41.8^\circ$$

RULES YOU NEED

$$\bullet \theta_{\text{crit}} = \sin^{-1}(n_2/n_1)$$

WORKING — STEP BY STEP**1 Identify the media**

Light starts in glass ($n_1 = 1.5$) and would go into air ($n_2 = 1$).

2 Substitute

$$\sin \theta_{\text{crit}} = 1 / 1.5 = 0.667$$

3 Take the inverse sine

$$\theta_{\text{crit}} = \sin^{-1}(0.667) = 41.8^\circ$$

Marks	Awarded for
1	Correct substitution $n_2/n_1 = 1/1.5$.
1	$\theta_{\text{crit}} \approx 41.8^\circ$.

6(B)(e)**[3 marks]**

A ray strikes the glass from air at $\theta_1 = 45^\circ$. Calculate the angle the light makes with the normal when it enters the water (θ_2). $n_{\text{water}} = 1.33$.

ANSWER

$\theta_2 \approx 32.1^\circ$ (angle in glass $\approx 28.1^\circ$)

RULES YOU NEED

- Snell's law at each boundary: $n_1 \sin \theta_1 = n_2 \sin \theta_2$
- Parallel faces: the angle leaving one face equals the angle hitting the next.

WORKING — STEP BY STEP**1 First boundary: air $\rightarrow$ glass**

$$1 \times \sin 45^\circ = 1.5 \times \sin \theta_{\text{glass}}$$

$$\sin \theta_{\text{glass}} = 0.7071 \div 1.5 = 0.4714 \Rightarrow \theta_{\text{glass}} = 28.1^\circ$$

It bends towards the normal ($45^\circ \rightarrow 28.1^\circ$), as expected going into denser glass.

2 Carry the angle across the glass

The two faces of the glass are parallel, so the ray hits the glass-water face at the same 28.1° to the normal.

3 Second boundary: glass $\rightarrow$ water

$$1.5 \times \sin 28.1^\circ = 1.33 \times \sin \theta_2$$

$$\sin \theta_2 = 1.5 \times 0.4714 \div 1.33 = 0.7071 \div 1.33 = 0.5317$$

4 Take the inverse sine

$$\theta_2 = \sin^{-1}(0.5317) = 32.1^\circ$$

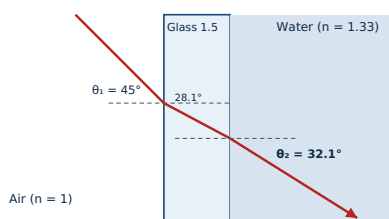


Fig. 4 — Bends towards the normal into glass, away from it into water.

5 Check with the shortcut

The glass terms cancel ($1.5 \times \sin \theta_{\text{glass}} = 1 \times \sin 45^\circ$), so directly: $\sin \theta_2 = \sin 45^\circ \div 1.33 = 0.5317 \rightarrow 32.1^\circ \checkmark$

Marks	Awarded for
1	Snell's law at the air-glass boundary: θ in glass $\approx 28.1^\circ$.
1	Uses the same angle at the second (parallel) face and applies Snell's law glass $\rightarrow$ water.
1	$\theta_2 \approx 32.1^\circ$.

Question 6(B) Total: 10 Marks